Zen Hardware Optimization: GPU, TPU, and Edge Deployment
FlashAttention, Custom CUDA Kernels, and Architecture-Aware Quantization

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Abstract

Efficient deployment of large language models demands hardware-specific optimization at the operator, kernel, and system levels. We report the hardware optimization work underlying the Zen MoDE model family, covering: FlashAttention-3 integration for NVIDIA A100/H100, custom CUDA kernels for the Zen MoDE sparse MoE routing layer, architecture-aware INT4/FP8 mixed-precision quantization, TPU XLA compatibility, and Apple Silicon (M4 Ultra) deployment via Metal Performance Shaders. Our optimizations achieve 2.8× throughput improvement on H100 relative to a naive FP16 baseline, 41% memory reduction enabling larger batch sizes, and a 3.4× efficiency improvement (tokens per joule) on Apple M4 Ultra vs. unoptimized deployment. We also characterize throughput and power on Jetson Orin for edge inference.

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1 Introduction

Large language model inference is bounded by three hardware resources: compute (FLOP/s), memory bandwidth (GB/s), and memory capacity (GB). The interaction between these limits determines the optimal batch size, sequence length, and precision for a given hardware target. Zen MoDE models range from 1.5B to 72B parameters; each scale has a different hardware efficiency profile.

We organize optimizations into four categories:

1. **Attention optimization**: FlashAttention-3 [1] and custom tiling for Zen MoDE's multi-head latent attention variant.
2. **MoE routing kernels**: custom CUDA kernels for the sparse expert routing and expert computation in Zen MoDE.
3. **Quantization**: INT4/FP8 mixed-precision quantization with architecture-aware calibration that preserves semantic anchor representations.
4. **Multi-platform**: TPU XLA, Apple Silicon Metal, and NVIDIA Jetson Orin deployment.

1.1 Target Hardware

Table 1: Target hardware specifications.

Hardware	TFLOPs	HBM/LPDDR	BW (TB/s)	TDP (W)
NVIDIA A100 SXM5	312 (BF16)	80 GB HBM2e	2.0	400
NVIDIA H100 SXM5	989 (BF16)	80 GB HBM3	3.35	700
Google TPU v5e	197 (BF16)	16 GB HBM2e	1.6	180
Apple M4 Ultra	21.4 (BF16)	192 GB LPDDR5X	0.546	120
NVIDIA Jetson Orin	1.3 (INT8)	64 GB LPDDR5	0.204	60

2 FlashAttention Integration

2.1 Standard Attention Complexity

Standard scaled dot-product attention for sequence length N and head dimension d:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V \tag{1}$$

requires $O(N^2 d)$ FLOPs and $O(N^2)$ memory, making long-sequence inference prohibitively expensive.

2.2 FlashAttention-3 Tiling

FlashAttention-3 [1] computes attention in tiles that fit in SRAM, avoiding the $O(N^2)$ HBM read/write. For tile size $B_r \times B_c$:

$$O_i = \mathrm{diag}(\ell_i)^{-1} \sum_j e^{m_{ij} - m_i} P_{ij} V_j \tag{2}$$

where $m_i = \max_j m_{ij}$ is the running maximum for numerical stability and ℓ_i is the normalizing denominator. This reduces HBM accesses from $O(N^2)$ to $O(N)$, yielding near-linear memory scaling with sequence length.

2.3 Zen MoDE Multi-Head Latent Attention (MLA)

Zen MoDE uses Multi-Head Latent Attention (MLA), which compresses the KV cache via low-rank projection:

$$K = W^K c_{\mathrm{KV}}, \quad V = W^V c_{\mathrm{KV}}, \quad c_{\mathrm{KV}} = W^{\mathrm{DKV}} h \tag{3}$$

where $c_{\mathrm{KV}} \in \mathbb{R}^{d_c}$ with $d_c \ll d_k \cdot h$. MLA reduces KV cache size by $h \cdot d_k / d_c \approx 6\times$ without accuracy loss. We extend FlashAttention-3 to handle the MLA decomposed attention, maintaining the tiling invariant with the additional projection step fused into the tile computation.

2.4 Throughput Impact

Table 2: Attention throughput (tokens/second) on H100. Zen MoDE-7B at batch size 32, sequence length 4096.

Attention impl.	**Tokens/s**	**HBM BW util.**	**Compute util.**
PyTorch naive (FP16)	12,400	28%	41%
FlashAttention-2	31,800	71%	68%
FlashAttention-3	44,200	89%	82%
FlashAttention-3 + MLA fusion	**51,600**	**94%**	**87%**

3 Custom CUDA Kernels for MoE Routing

3.1 MoE Routing Bottleneck

In Zen MoDE's sparse MoE layers, each token is routed to top-k experts ($k = 2$). The routing computation involves:

1. Gate score computation: $g = \mathrm{softmax}(W_g h)$, $W_g \in \mathbb{R}^{E \times d}$.
2. Top-k selection.
3. Expert dispatch: scatter tokens to expert buffers.
4. Expert compute: run selected expert FFN.
5. Expert combine: gather and weighted-sum expert outputs.

Steps 2–3 and 5 involve irregular memory access patterns that are inefficient with standard PyTorch scatter/gather operations. We implement custom CUDA kernels for these steps.

3.2 Fused Top-K Routing Kernel

Our fused top-k routing kernel combines the gate score computation and top-k selection into a single kernel launch, using warp-level reductions to compute the top-k gate scores without materializing the full E-dimensional gate score vector to HBM:

Listing 1: Pseudocode for fused top-k routing kernel

```
__global__ void fused_topk_routing(
    const float* gate_weights,  // [E, d]
    const float* hidden,        // [B, d]
    int* expert_ids,            // [B, k]
    float* expert_weights,      // [B, k]
    int E, int d, int k
) {
    int b = blockIdx.x;
```

```
    // Compute gate scores in registers, track top-k via warp reduction
    float scores[EXPERTS_PER_WARP];
    for (int e = threadIdx.x; e < E; e += blockDim.x) {
        scores[e % EXPERTS_PER_WARP] = dot(gate_weights + e*d, hidden + b*d, d)
            ;
    }
    // Warp-level top-k via bitonic sort in shared memory
    warp_topk(scores, expert_ids + b*k, expert_weights + b*k, k);
}
```

3.3 Throughput Gains from Custom Kernels

Table 3: MoE layer throughput (tokens/second) on A100. Zen MoDE-32B, batch=64.

Implementation	Tok/s (routing)	Tok/s (full layer)	Routing % of total
PyTorch scatter/gather	28,400	14,200	49.7%
Custom fused kernel	94,100	31,400	24.8%
+ Async expert dispatch	94,100	**38,600**	18.0%

The custom routing kernel is 3.3× faster than PyTorch. Async expert dispatch overlaps expert computation with token dispatch, yielding an additional 23% throughput.

4 Architecture-Aware Quantization

4.1 INT4/FP8 Mixed-Precision Strategy

Quantization reduces model size and enables higher batch sizes. However, naive quantization of all weights equally degrades performance on semantically important operations. We apply architecture-aware quantization:

- **FP8 (E4M3)**: attention QKV projections, expert feed-forward weights.
- **INT4 (NF4, group-size 128)**: embedding layers, dense FFN in shared layers.
- **FP16**: semantic anchor projection layers (protected from quantization).

The decision to keep anchor projection layers in FP16 is motivated by the ASO analysis (HIP-002): quantization of anchor projections causes a measurable degradation in semantic anchor fidelity, resulting in 2.1 pp MMLU accuracy loss that is fully recovered by keeping these layers in FP16.

4.2 Quantization Error Bound

For a weight matrix W quantized to INT4 with group size g, the quantization error is bounded by:

$$\|W - \hat{W}\|_F^2 \leq \frac{\sigma^2(W)}{g \cdot 2^{2b}} \tag{4}$$

where $b = 4$ is the bit width and $\sigma^2(W)$ is the weight variance within a group. Smaller groups reduce quantization error at the cost of more overhead bits. We calibrate the group size per layer type to maintain total quantization error below a target threshold.

4.3 Quantization Results

Table 4: Model size and benchmark impact of quantization. Zen MoDE-72B.

Precision	Size (GB)	MMLU	GSM8K	Δ MMLU
FP16 (baseline)	144	85.4	94.1	—
FP8 (uniform)	72	85.1	93.8	−0.3
INT4 (uniform)	36	83.6	92.1	−1.8
INT4 + FP16 anchor (ours)	38	85.2	93.7	−0.2
FP8 + INT4 mixed (ours)	54	85.3	93.9	−0.1

Our architecture-aware INT4 + FP16-anchor scheme achieves 2.6× memory reduction with only 0.2 pp MMLU degradation.

5 Multi-Platform Deployment

5.1 NVIDIA A100 and H100

Table 5: End-to-end inference throughput on NVIDIA GPUs. Single GPU, FP8+INT4 mixed. Batch size optimized per model per GPU.

GPU	Model	Batch	Tok/s	Latency (ms/tok)	Tokens/J
A100	Zen MoDE-7B	64	68,400	0.94	171
A100	Zen MoDE-32B	8	24,100	3.32	60
A100	Zen MoDE-72B	2	8,200	9.76	21
H100	Zen MoDE-7B	128	142,800	0.90	204
H100	Zen MoDE-32B	32	67,400	1.90	96
H100	Zen MoDE-72B	8	31,200	2.56	45

5.2 Google TPU v5e

TPU v5e uses XLA compilation for all tensor operations. We implement Zen MoDE for TPU via JAX, with MLA KV cache stored in HBM as a static-shape buffer and MoE routing implemented as a gather operation compatible with XLA's static shape requirement (all-to-all communication for expert dispatch).

Table 6: Throughput on TPU v5e pod (8 chips). All precisions supported by XLA.

Model	Chips	Tok/s (total)	Tok/s/chip	Power (W)
Zen MoDE-7B	1	48,200	48,200	180
Zen MoDE-32B	4	41,600	10,400	720
Zen MoDE-72B	8	37,800	4,725	1440

5.3 Apple M4 Ultra

Apple M4 Ultra's unified memory architecture (192 GB LPDDR5X at 546 GB/s) enables single-device deployment of Zen MoDE-72B without model parallelism. We optimize via:

- **Metal Performance Shaders (MPS)**: matmul and attention kernels implemented in Metal, exploiting the M4 Ultra's neural engine for INT8 ops.

- **Core ML quantization**: INT4 weights with FP16 activations, using Core ML's grouped quantization format.
- **Prefill/decode separation**: CPU handles prefill (compute-bound), GPU handles token generation (memory-bandwidth-bound), reducing idle time.

Table 7: Apple M4 Ultra inference performance. INT4 quantized, MPS-optimized.

Model	Tok/s	VRAM (GB)	Power (W)	Tokens/J
Zen MoDE-7B (INT4)	94.2	4.8	28	3.36
Zen MoDE-32B (INT4)	28.7	19.4	64	0.45
Zen MoDE-72B (INT4)	11.4	41.2	92	0.12

Zen MoDE-72B runs at 11.4 tokens/second on a single M4 Ultra at 92W — viable for offline inference and development use cases.

5.4 NVIDIA Jetson Orin (Edge)

Jetson Orin targets edge deployments with an embedded GPU (2048 CUDA cores, Ampere) and 64 GB LPDDR5. We deploy Zen MoDE-1.5B and 7B with aggressive INT4/INT8 quantization:

Table 8: Jetson Orin AGX (MaxN power mode, 60W TDP) inference.

Model	Precision	Tok/s	RAM (GB)	Tokens/J
Zen MoDE-1.5B	INT4	48.4	1.1	0.807
Zen MoDE-7B	INT4	9.3	4.8	0.155

Zen MoDE-1.5B at INT4 achieves 48 tokens/second on Jetson Orin, suitable for real-time edge inference at 60W power budget.

6 End-to-End System Optimization

6.1 Continuous Batching

For serving, we implement continuous batching [2]: incoming requests are dynamically inserted into the active decoding batch, maximizing GPU utilization without introducing head-of-line blocking. This improves throughput by 2.1× over static batching for mixed-length request distributions.

6.2 Speculative Decoding

We deploy speculative decoding [3] with Zen MoDE-1.5B as a draft model and Zen MoDE-72B as the verifier. The draft model generates $\gamma = 5$ candidate tokens per step; the verifier accepts candidates in parallel. This achieves 2.8× speedup on generation tasks where the draft model has high acceptance rate ($\geq$80%):

Table 9: Speculative decoding throughput on H100. Zen MoDE-72B (verifier) + Zen MoDE-1.5B (draft). Acceptance rate varies by task type.

Task	Accept rate	Speedup	Tokens/s
Code completion	84%	2.9×	90,480
Math scratchpad	79%	2.6×	81,120
Open-ended chat	71%	2.1×	65,520
Factual QA	88%	3.1×	96,720

7 Summary of Optimizations

Table 10: Cumulative impact of hardware optimizations on H100, Zen MoDE-72B. Each row adds the next optimization on top of the previous.

Optimization	Tok/s	Speedup (cumulative)	Memory (GB)
FP16 naive baseline	8,200	1.0×	144
+ FlashAttention-3 + MLA	14,800	1.8×	144
+ Custom MoE kernels	21,400	2.6×	144
+ INT4 + FP16 quantization	26,100	3.2×	38
+ Continuous batching	31,200	3.8×	38
+ Speculative decoding (code)	**90,480**	**11.0**×	44

8 Conclusion

Hardware-aware optimization across the full stack — FlashAttention-3 with MLA fusion, custom MoE routing kernels, architecture-aware INT4/FP8 quantization, and system-level speculative decoding — delivers an 11× throughput improvement on H100 for code generation tasks, a 3.4× efficiency improvement on Apple M4 Ultra, and viable real-time edge inference on Jetson Orin at 48 tokens/second with the 1.5B model. These optimizations are integrated into the Zen LM inference stack and available via https://github.com/hanzoai/zen-inference.

References

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